INCH-POUND

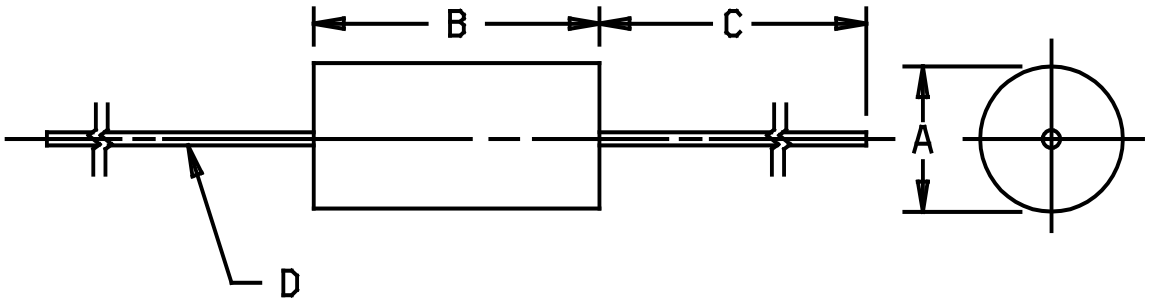
MIL-PRF-39010/5E
W/AMENDMENT 2
14 December 2022
SUPERSEDING
MIL-PRF-39010/5E
W/AMENDMENT 1
6 September 2018

PERFORMANCE SPECIFICATION SHEET

COILS, FIXED, RADIO FREQUENCY, MOLDED, SUBMINIATURE (POWDERED IRON CORE), ESTABLISHED RELIABILITY AND NONESTABLISHED RELIABILITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39010.



Letter	Dimensions are in inches with metric equivalents (mm) in parentheses	Dimensions are in inches with metric equivalents (mm) in parentheses
	Minimum	Maximum
A	.157 (3.99)	.219 (5.56)
B	.407 (10.34)	.469 (11.91)
C	1.250 (31.75)	1.626 (41.30)
D	.023 (0.58)	.027 (0.69)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.

FIGURE 1. Dimensions and configuration.

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I
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FSC 5950

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REQUIREMENTS:

Interface and physical dimensions: See figure 1.

Material: Powdered iron core.

Weight: 0.03 ounce maximum.

Operating temperature range: -55°C to +105°C.

Dielectric withstanding voltage: MIL-STD-202-301; test voltage of 1,000 V rms.

Barometric pressure: MIL-STD-202-105, test condition C, 70,000 feet with a test voltage of 200 V rms.

Electrical characteristics: See table I and table II.

Inductance: See table I.

Inductance tolerance: See table I.

Q values: See table I.

Self-resonant frequency: See table I.

DC resistance: See table I.

Temperature rise: 15°C.

Terminal pull: 5 pounds.

Part or Identifying Number (PIN): M39010/05 (dash number from table I).

Supersession data: This specification supersedes MS75101.

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TABLE I. Electrical characteristics (initial) and dash numbers.

Dash number 1/	Inductance μH	Inductance tolerance ± percent	Q minimum	Test frequency (MHz)	Self-resonant frequency minimum (MHz)	DC resistance (25°C) maximum (ohms)	Rated dc current (mA) 2/
A3R0**	3.00	5	30	7.9	70	0.140	945
A3R3**	3.30	5, 10	30	7.9	70	0.140	945
A3R6**	3.60	5	30	7.9	65	0.155	870
A3R9**	3.90	5, 10	30	7.9	65	0.155	870
A4R3**	4.30	5	30	7.9	60	0.210	745
A4R7**	4.70	5, 10	30	7.9	60	0.210	745
A5R1**	5.10	5	30	7.9	50	0.280	645
A5R6**	5.60	5, 10	30	7.9	50	0.280	645
A6R2**	6.20	5	30	7.9	50	0.375	560
A6R8**	6.80	5, 10	30	7.9	50	0.375	560
A7R5**	7.50	5	30	7.9	48	0.440	520
A8R2**	8.20	5, 10	30	7.9	48	0.440	520
A9R1**	9.10	5	30	7.9	42	0.605	440
A100**	10.0	5, 10	30	7.9	42	0.605	440
A110**	11.0	5	30	2.5	36	1.05	335
A120**	12.0	5, 10	50	2.5	36	.05	335
A130**	13.0	5	50	2.5	30	1.20	310
A150**	15.0	5, 10	55	2.5	30	1.20	310
A160**	16.0	5	55	2.5	30	1.95	245
A180**	18.0	5, 10	60	2.5	30	1.95	245
A200**	20.0	5	60	2.5	24	2.20	230
A220**	22.0	5, 10	60	2.5	24	2.20	230
A240**	24.0	5	60	2.5	22	2.75	205
A270**	27.0	5, 10	65	2.5	22	2.75	205

1/ The complete dash number will include two additional letters (indicated by **). The first additional letter will indicate the inductance tolerance and the second additional letter will indicate the product level (e.g., C, M, P, R, S) and will be added to the end of the dash number.

2/ The rated dc current is based on 90°C ambient temperature with a 15°C rise.

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TABLE II. Electrical characteristics (final). 1/

Inspection group	Allowable variation from initial measurement	Allowable variation from initial measurement	Allowable percent from specified minimum value in electrical characteristics (initial) table	Allowable percent from specified minimum value in electrical characteristics (initial) table
	Inductance (percent)	DC resistance	Self-resonant frequency	Q
Qualification inspection				
Group II	±2			-10
Group IV	±5	±(3% +.001ohm)	2/	-10
Group V	±5	±(2% +.001ohm)	2/	-15
Quality conformance inspection Group B				
Subgroup 1	±5	±(2% +.001ohm)	2/	-15
Subgroup 3	±2			-10
Subgroup 4	±5	±(3% +.001ohm)	2/	-10

1/ Test fixture allowance of +.01 μH shall be added to all change in inductance limits ±(_percent +.01 μH).
2/ The self-resonant frequency shall be not less than the value specified in table I.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-39010, this document references the following:

MIL-STD-202-105, MIL-STD-202-301

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Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA – CC

Preparing activity:
DLA - CC

Review activities:
Army - AR, CR4, MI
Navy - AS, CG, MC, OS, SH
Air Force - 19

(Project 5950-2022-073)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at https://assist.dla.mil.

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